

Making molecules

The naturally occurring elements are the building blocks of everything in our world. The atoms of various elements can be joined in a wide variety of ways to produce many compounds. Elements and compounds can be combined in many ways to make countless mixtures.

Atoms can join, or bond, in many different ways. In some substances, atoms are joined in groups called **molecules**. For example, in oxygen gas, oxygen atoms are joined in groups of two. In the

compound carbon dioxide, one carbon and two oxygen atoms are joined in every molecule. Atoms can join to form small or large molecules of many different shapes.

Some compounds are not made up of molecules. Instead the atoms bond by lining up one after the other. Sodium bonds to chlorine, which bonds to sodium and so on. Common table salt is an example of a substance that is bonded in this way.

INVESTIGATION 7.8

Mix 'n' match

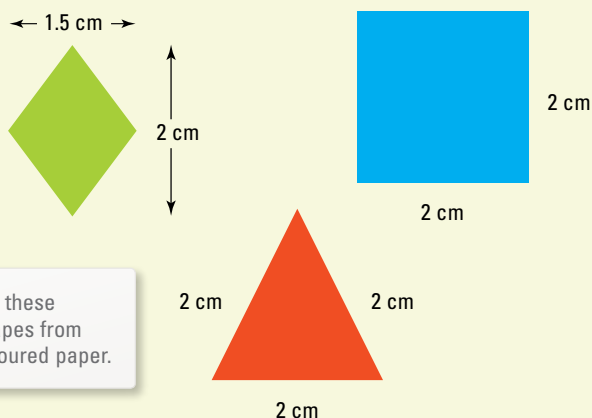
AIM To model the molecules of a variety of compounds

Materials:

green, red and blue sheets of paper
scissors pencil ruler
1 large sheet of cartridge paper

METHOD AND RESULTS

- ▶ Cut out 15 diamonds, each 2 cm long and 1.5 cm wide, from the green sheet of paper.
- ▶ Cut out 30 equilateral triangles, with each side 2 cm, from the red sheet of paper.
- ▶ Cut out 15 squares, with each side 2 cm, from the blue sheet of paper.



Cut these shapes from coloured paper.

- ▶ Imagine that different types of atoms are represented by particular shapes:
a blue square = carbon
a green diamond = oxygen
a red triangle = hydrogen
and that, by placing them side by side on the sheet of paper, you are joining them.

- ▶ Place two green diamonds next to each other on the sheet. This represents the oxygen molecule, as shown in the diagram below.



A green diamond represents an atom of oxygen. Together, two diamonds represent a molecule of oxygen.

- ▶ Place one blue square on the sheet between two green diamonds. This represents the compound carbon dioxide. Label it with its name and symbol.
- ▶ Represent and label the following substances:
 - water, which contains 1 oxygen and 2 hydrogen atoms
 - methane (natural gas), which contains 1 carbon and 4 hydrogen atoms
 - benzene (in petrol), which contains 6 carbon and 6 hydrogen atoms
 - glucose (sugar), which contains 6 carbon, 12 hydrogen and 6 oxygen atoms
 - hydrogen peroxide (found in hair bleach), which contains 2 oxygen atoms and 2 hydrogen atoms.

DISCUSS AND EXPLAIN

- 1 Which of these compounds contain only hydrogen and carbon atoms?
- 2 In what ways are these two substances different from each other?
- 3 Which of the compounds contain only oxygen and hydrogen? Do these compounds have the same characteristics?
- 4 Think about the appearance of the compound sugar. How does it differ in appearance from the elements from which it is made?